Divas Subedi

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Education

M.S. in Computer Science
Jan 2025 - May 2027 (Expected)

Georgia Institute of Technology — *Atlanta, GA, USA*

- Specialization: Machine Learning
- Current GPA: 4.0
- Current Research Project: Systematic Literature Review of Machine Learning Applications in Exoplanetary Science
- Course Project: TORTOISE, fine-tuned SAM 2 for segmentation of mining sites in satellite imagery

Relevant Coursework

- Machine Learning
- Deep Learning
- Intro to Research
- Network Science (Applied Graph Theory)
- Intro to HPC†
- Bayesian Statistics†

†In progress, Fall 2026

B.S. in Physics and Computer Engineering
Jan 2019 - May 2022

Trinity College — *Hartford, CT, USA*

- Cumulative GPA: 4.1
- Physics Senior Project: Analysis of BB84 Quantum Key Distribution Algorithm
- Engineering Senior Project: Biometric Signature Authentication

Relevant Coursework

- Relativity and Fundamental Particles
- Digital Signal Processing
- Numerical Approach to PDE

Tech Skills

Programming Python {Qiskit, QuTiP, SciPy, Scikit-learn, PyTorch}, MATLAB, C++, C, Mathematica
AI/ML Neural Networks, Transformers, Foundation Model Fine-Tuning {SAM 2}, Feature Engineering {Spectral, MFCC}
Software Linux, Vitis, Vivado, Slurm, Geant4 {G4CMP}, COMSOL, SPICE
Hardware SoC FPGA (Zynq), Jetson Nano

Internship Projects

Quantum Sensor for detection of Dark Matter [T2][P5]
May 2022 - Aug 2022

Fermi National Accelerator Laboratory, PPD (SIST Internship) — *Batavia, IL, USA*

- Used Geant4 (G4CMP) to simulate phonon generation and transport in a silicon substrate to support calibration of an MKID-based dark matter detector.
- Ran parameter sweeps over interaction energy, position, and momentum and analyzed the output into phonon hit maps, collection times, and energy spectra at the electrode.
- Identified and documented a bug in how G4CMP distributed energy between charge-carrier pairs, and presented the findings to the library's maintainers.

Ground-induced Impedance Monitor for DUNE (GIZMo) [T1][P2]
May 2021 - Aug 2021

Fermi National Accelerator Laboratory, PPD (SIST Internship) — *Batavia, IL, USA*

- Contributed to the design and development of GIZMo, a real-time ground impedance monitor for the DUNE far detector.
- Implemented a digital lock-in amplifier using quadrature demodulation on a Zynq SoC-FPGA to extract impedance magnitude and phase from the digitized response signal.
- Used L-BFGS in Python to find optimal inductor values for the detector's parasitic capacitance via LTspice simulation.

Selected Research Projects

Biometric Signature Authentication [C5,C7][P4,P6]
Sep 2021 - Feb 2023

Trinity College Department of Engineering — *Hartford, CT, USA*

- Developed an accelerometer-integrated stylus that authenticates users by analyzing inertial data captured during signature.
- Built a high-speed data collection system using embedded sensors in C for real-time communication with the Raspberry Pi.
- Crafted manual features using spectral analysis of time-series data for classification using MATLAB.
- Implemented a neural network, using manually crafted features and MFCC, capable of authenticating users with 94% accuracy.

Vibration-Based Contact Sensing [C3,C4][P3]
May 2020 - Sep 2021

Trinity College Department of Engineering — *Hartford, CT, USA*

- Designed a low-cost vibration-based contact sensor for tactile sensing.
- Characterized the rod's motion via system identification and crafted manual features for classification.
- Implemented a neural network in MATLAB and Python to classify contact location using manually crafted features and MFCC with accuracy of 95%.

Semiconductor Device Modeling [C1][P1] *May 2019 - Aug 2019*

Trinity College Department of Engineering *Hartford, CT, USA*

- Used COMSOL Multiphysics to study the effects of constant-field scaling on FIBMOS compared to conventional MOSFETs in the nanometer regime.
- Designed simulation to reproduce and expand on work by Shen et al. (1998) on 122.5-nm to 350-nm channel-length devices.
- Analyzed results showing FIBMOS advantages in threshold voltage stability, output resistance, punch-through resistance, and mitigation of hot electron degradation compared to traditional MOSFETs.

Work Experience

Software Developer I *Sep 2022 - Present*

Fermi National Accelerator Laboratory, ITD *Batavia, IL, USA*

- Build and maintain secured ETL pipelines, a master data service, and EH&S databases behind laboratory dashboards and data exchange with other government agencies.
- Develop and maintain a data virtualization platform for real-time integration across separate data sources.
- Build LLM-based developer tools and an agent for navigating internal codebases and their documentation.

Publications

[J1] K. Huang, **D. Subedi**, R. Mitra, I. Yung, K. Boyd, E. Aldrich, and D. Chitrakar, "Telelocomotion—Remotely Operated Legged Robots", *Applied Sciences 2021*, vol. 11, no. 1:194.

[C7] **D. Subedi**, D. Chitrakar, I. Yung, Y. Zhu, Y. Su, K. Huang, "Biometric Signature Authentication with Low Cost Embedded Stylus", *2023 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, Seattle, WA, USA, 2023.

[C6] **D. Subedi**, W. Jiang, R. Rahman, H. Zhang, K. Huang, Y. Su, "Smoothness Constrained Curiosity Driven Multicamera Trajectory Optimization for Robot-Assisted Minimally Invasive Surgery", *2023 International Symposium on Medical Robotics (ISMR)*, Atlanta, GA, USA, 2023.

[C5] **D. Subedi**, I. Yung, D. Chitrakar, K. Huang, "Inertial-Measurement-Based Biometric Authentication of Handwritten Signature", *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, Glasgow, Scotland, United Kingdom, 2022.

[C4] **D. Subedi**, E. Schoemer, D. Chitrakar, Y. Su and K. Huang, "Contact Location via Active Oscillatory Actuation", *2022 IEEE/SICE International Symposium on System Integration (SII)*, Narvik, Norway, 2022.

[C3] R. Mitra, K. Boyd, **D. Subedi**, D. Chitrakar, E. Aldrich, A. Swamy, and K. Huang, "Contact Sensing via Active Oscillatory Actuation", *2020 3rd International Conference on Mechatronics, Robotics and Automation (ICMRA)*, Shanghai, China, 2020.

[C2] K. Huang, D. Chitrakar, R. Mitra, **D. Subedi**, and Y. Su, "Characterizing Limits of Vision-Based Force Feedback in Simulated Surgical Tool-Tissue Interaction", *2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, Montreal, QC, Canada, 2020.

[C1] **D. Subedi** and D. A. Fixel, "MOSFET Channel Engineering and Scaling Study using COMSOL® Multiphysics Simulation Software", *2019 COMSOL Conference*, Boston, MA, USA, 2019.

Talks & Presentations

[P6] "Biometric Signature Authentication with Low Cost Embedded Stylus", *2023 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, Seattle, WA, USA, June 29, 2023.

[P5] "Quantum Sensor for detection of Dark Matter", *Fermilab Summer Internships in Science & Technology (SIST)*, Batavia, IL, August 12, 2022.

[P4] "Inertial-Measurement-Based Biometric Authentication of Handwritten Signature", *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, Glasgow, Scotland, United Kingdom, July 14, 2022.

[P3] "Contact Location via Active Oscillatory Actuation", *2022 IEEE/SICE International Symposium on System Integration (SII)*, Narvik, Norway, January 9, 2022.

[P2] "GIZMo for DUNE at LBNF", *Fermilab Summer Internships in Science & Technology (SIST)*, Batavia, IL, August 9, 2021.

[P1] "MOSFET Channel Engineering and Scaling Study using COMSOL Multiphysics Simulation Software", *2019 COMSOL Conference*, Boston, MA, USA, October 2, 2019.

Technical Reports

[T2] **D. Subedi**, K. Stifter, "Quantum Sensor for detection of Dark Matter", *Fermilab Summer Internships in Science & Technology (SIST)*, Batavia, IL, 2022.

[T1] **D. Subedi**, M.J. Utes, P.M. Rubinov, "GIZMo for DUNE at LBNF", *Fermilab Summer Internships in Science & Technology (SIST)*, Batavia, IL, 2021.